

RE and FSM - Results



Attempt 1 of 1

Written Jan 15, 2026 10:14 AM - Jan 15, 2026 10:52 AM

Attempt Score **10 / 10 - 100 %**

Overall Grade (Highest Attempt) **10 / 10 - 100 %**

Question 1

1 / 1 point

Which language is not regular?

- ☐ $\{ w \in \{0,1\}^* \mid w \text{ ends with } 01 \}$
- ☐ $\{ w \in \{a,b\}^* \mid \text{number of } a\text{'s is even} \}$
- ☒ $\{ a^n b^n \mid n \geq 0 \}$
- ☐ $\{ w \in \{0,1\}^* \mid w \text{ contains substring } 101 \}$

Question 2

1 / 1 point

If two regular expressions describe the same language, then:

- ☐ They must be syntactically identical
- ☒ They must be equivalent

- ☐ They must generate the same DFA states
- ☐ They must use the same operators

Question 3

1 / 1 point

Which RE describes the same language as $(a^*)^*$?

- ☐ ϵ
- ☐ $(aa)^*$
- ☐ a^+
- ☒ a^*

Question 4

1 / 1 point

Which pair of regular expressions are not equivalent?

- ☐ $(a|b)^*$ and $(b|a)^*$
- ☒ a^* and $(aa)^*$
- ☐ $(ab)^*$ and $a(ba)^*b \mid \epsilon$
- ☐ $(a^*)^*$ and a^*

Question 5

1 / 1 point

Given two regular expressions r and s , which of the following is always true?

- ☐ $rs = sr$
- ✓ ☒ $r|s=s|r$
- ☐ $r^* | s^* = (r | s)^*$
- ☐ $r^*s = r s^*$

Question 6

1 / 1 point

Given three regular expressions r, s, t , which of the following is valid?

- ✓ ☒ $(rs)t=r(st)$
- ☐ $(r|s) t = r | (st)$
- ☐ $(r|s) t = r (s|t)$
- ☐ $rst = tsr$

Question 7

1 / 1 point

Regular expression $(s|t) r$ is equal to

- ☐ $st | r$
- ☐ $sr | rt$
- ☐ $s(t|r)$
- ✓ ☒ $tr | sr$

Question 8

1 / 1 point

Which feature makes an NFA different from a DFA?

- ☐ It can recognize more languages
- ✓ ☒ It allows ϵ -transitions

- ☐ It always has fewer states
- ☐ It cannot be converted to a DFA

Question 9

1 / 1 point

Which statement is true?

- ☐ Not every NFA has an equivalent DFA
- ✓ ☒ DFA is a special case of NFA
- ☐ NFAs accept fewer languages than DFAs
- ☐ NFAs are more powerful than DFAs

Question 10

1 / 1 point

Which conversion is always possible?

- ☐ DFA $\rightarrow$ NFA
- ☐ NFA $\rightarrow$ DFA
- ☐ RE $\rightarrow$ DFA
- ✓ ☒ All of the above

Done